



First semester exam + correction

Exercise 1

- 1 Let $z = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$
 - a Find the square root of z .
 - b Write z in the polar form.
 - c Deduce the exact value of $\cos(\frac{\pi}{8})$ and $\sin(\frac{\pi}{8})$.
- 2 Using the definition of the limit, show that $\lim_{n \rightarrow +\infty} 1 + \frac{(-1)^n}{n} = 1$.
- 3 Prove that $\forall x \in]-1, 1[, (\arcsin(x))' = \frac{1}{\sqrt{1-x^2}}$.

Solution 1

1 Let $z = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$

- a Find the square root of z(01 pt)

$$\text{Let } w = a + ib \text{ s.t. } w^2 = z \implies a^2 - b^2 + 2iab = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$$

$$\text{and } |w|^2 = |z| \implies a^2 + b^2 = 1 \text{ so } \begin{cases} a^2 - b^2 = \frac{\sqrt{2}}{2} & (1) \\ a^2 + b^2 = 1 & (2) \\ 2ab = \frac{\sqrt{2}}{2} & (3) \end{cases}$$

$$(1) + (2) \implies 2a^2 = 1 + \frac{\sqrt{2}}{2} = \frac{2 + \sqrt{2}}{2} \implies a^2 = \frac{2 + \sqrt{2}}{4}$$

$$\text{so } a = \frac{\sqrt{2 + \sqrt{2}}}{2} \text{ or } a = -\frac{\sqrt{2 + \sqrt{2}}}{2}$$

$$\text{and } (2) - (1) \implies 2b^2 = 1 - \frac{\sqrt{2}}{2} = \frac{2 - \sqrt{2}}{2} \implies b^2 = \frac{2 - \sqrt{2}}{4}$$

$$\text{so } b = \frac{\sqrt{2 - \sqrt{2}}}{2} \text{ or } b = -\frac{\sqrt{2 - \sqrt{2}}}{2}$$

$$\text{From } (3), a \text{ and } b \text{ have the same sign so } w_1 = \frac{\sqrt{2 + \sqrt{2}}}{2} + i\frac{\sqrt{2 - \sqrt{2}}}{2}$$

$$\text{and } w_2 = -\frac{\sqrt{2 + \sqrt{2}}}{2} - i\frac{\sqrt{2 - \sqrt{2}}}{2}$$

b The polar form of z(0.5 pt)

$$\text{We have } r = \sqrt{x^2 + y^2} = \sqrt{\frac{2}{4} + \frac{2}{4}} = 1 \quad \text{and} \quad \begin{cases} \cos(\theta) = \frac{x}{r} = \frac{\sqrt{2}}{2} \\ \sin(\theta) = \frac{y}{r} = \frac{\sqrt{2}}{2} \end{cases} \quad \text{so } \theta = \frac{\pi}{4}$$

$$\text{Therefore } z = r(\cos(\theta) + i \sin(\theta)) \implies z = \cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right)$$

c The exact value of $\cos(\frac{\pi}{8})$ and $\sin(\frac{\pi}{8})$ (01 pt)

The square roots of z are:

$$w_k = e^{i(\frac{\pi}{4} + 2k\pi)} \quad ; \quad k = 0, 1$$

$$w_0 = e^{i\frac{\pi}{8}} = \cos\left(\frac{\pi}{8}\right) + i \sin\left(\frac{\pi}{8}\right) \quad \text{and} \quad w_1 = e^{i\frac{\pi}{8} + \pi} = \cos\left(\frac{9\pi}{8}\right) + i \sin\left(\frac{9\pi}{8}\right)$$

$$\text{and because } \cos\left(\frac{\pi}{8}\right) > 0, \sin\left(\frac{\pi}{8}\right) > 0$$

$$\text{we get } \cos\left(\frac{\pi}{8}\right) + i \sin\left(\frac{\pi}{8}\right) = \frac{\sqrt{2+\sqrt{2}}}{2} + i \frac{\sqrt{2-\sqrt{2}}}{2}$$

$$\text{so } \begin{cases} \cos\left(\frac{\pi}{8}\right) = \frac{\sqrt{2+\sqrt{2}}}{2} \\ \sin\left(\frac{\pi}{8}\right) = \frac{\sqrt{2-\sqrt{2}}}{2} \end{cases}$$

2 Using the definition of the limit, show that $\lim_{n \rightarrow +\infty} 1 + \frac{(-1)^n}{n} = 1$(01 pt)

$$\text{We have } \lim_{n \rightarrow \infty} u_n = l \iff \forall \epsilon > 0; \exists n_0 \in \mathbb{N}; \quad n \geq n_0 \implies |u_n - l| \leq \epsilon$$

$$\text{Let } \epsilon > 0; \quad |u_n - l| = \left| 1 + \frac{(-1)^n}{n} - 1 \right| = \left| \frac{(-1)^n}{n} \right| = \frac{1}{n} \leq \epsilon \implies n \geq \frac{1}{\epsilon}$$

$$\text{we take } n_0 = \left\lceil \frac{1}{\epsilon} \right\rceil + 1 \in \mathbb{N}; \quad \forall n \in \mathbb{N}; \quad n \geq n_0 \implies |u_n - 1| \leq \epsilon$$

3 Prove that $\forall x \in]-1, 1[, (\arcsin(x))' = \frac{1}{\sqrt{1-x^2}}$ (01,5 pt)

We have $(f^{-1}(x))' = \frac{1}{f'(f^{-1}(x))}$

Let $f(x) = \sin(x)$

* f is differentiable on $]-\frac{\pi}{2}, \frac{\pi}{2}[$

* $\forall x \in]-\frac{\pi}{2}, \frac{\pi}{2}[; (\sin x)' = \cos x \neq 0$

so $f^{-1} = \arcsin$ is differentiable on $]-\frac{\pi}{2}, \frac{\pi}{2}[$ and $\forall x \in]-1, 1[;$

$$(\arcsin(x))' = \frac{1}{\cos(\arcsin x)}$$

Let $x \in]-1, 1[$ and $y = \arcsin x \implies y \in]-\frac{\pi}{2}, \frac{\pi}{2}[\implies \cos y > 0$

Since $\cos^2(y) + \sin^2(y) = 1 \implies \cos(y) = \sqrt{1 - \sin^2(y)}$

$$\implies \cos(\arcsin x) = \sqrt{1 - \sin^2(\arcsin x)} = \sqrt{1 - x^2}$$

We obtain $\forall x \in]-1, 1[, (\arcsin(x))' = \frac{1}{\sqrt{1-x^2}}$

Exercise 2

1 State the mean value theorem.

2 Let $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} = \sum_{k=1}^n \frac{1}{k}; n \geq 1$.

- a** Using the mean value theorem, prove that: $\frac{1}{x+1} \leq \ln(x+1) - \ln(x) \leq \frac{1}{x}, \forall x \in \mathbb{R}_+^*$.
- b** Deduce that $\ln(n+1) \leq H_n \leq \ln(n) + 1, \forall n \geq 1$.
- c** Determine the limit of H_n .
- d** Let $u_n = H_n - \ln(n), \forall n \geq 1$.
Prove that (u_n) is decreasing and positive. What can you deduce?

3 Let's consider the sequence (v_n) defined by $v_0 = \frac{3}{2}$ and $v_{n+1} = \sqrt{3v_n - 2}$ for $n \geq 1$.

- a** Show that: $1 \leq v_n \leq 2, \forall n \in \mathbb{N}$.
- b** Study the monotonicity of (v_n) and determine its limit.

Solution 2

1 The mean value theorem(01 pt)

Let f be a function defined on $[a, b]$, if we have

- * f is continuous on $[a, b]$
- * f is differentiable on $]a, b[$

then there exists $c \in]a, b[; \frac{f(b) - f(a)}{b - a} = f'(c)$

2 Let $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} = \sum_{k=1}^n \frac{1}{k}; n \geq 1$.

a *Prove that:* $\frac{1}{x+1} \leq \ln(x+1) - \ln(x) \leq \frac{1}{x}, \forall x \in \mathbb{R}_+^* \dots \dots \dots (01 \text{ pt})$

Let $f(x) = \ln(x)$ for $x > 0$; f is defined on $[x, x+1]$

- * f is continuous on $[x, x+1]$
- * f is differentiable on $]x, x+1[$

Then $\exists c \in]x, x+1[; \frac{f(x+1) - f(x)}{x+1 - x} = f'(c) = \frac{1}{c} \implies \ln(x+1) - \ln(x) = \frac{1}{c}$

we have $x < c < x+1 \implies \frac{1}{x+1} < \frac{1}{c} < \frac{1}{x}$

so $\frac{1}{x+1} < \ln(x+1) - \ln(x) < \frac{1}{x}; \quad \forall x > 0$

b *Deduce that* $\ln(n+1) \leq H_n \leq \ln(n) + 1, \forall n \geq 1 \dots \dots \dots (01,5 \text{ pt})$

According to (2.a) $\frac{1}{n+1} \leq \ln(n+1) - \ln(n) \leq \frac{1}{n}; \quad n \geq 0$

so for

$$\begin{array}{llll} n = 1; & \frac{1}{2} & \leq & \ln(2) - \ln(1) & \leq & 1 \\ n = 2; & \frac{1}{3} & \leq & \ln(3) - \ln(2) & \leq & \frac{1}{2} \\ n = 3; & \frac{1}{4} & \leq & \ln(4) - \ln(3) & \leq & \frac{1}{3} \\ \dots & ; & \dots & \leq & \dots & \leq \dots \\ n - 1; & \frac{1}{n} & \leq & \ln(n) - \ln(n-1) & \leq & \frac{1}{n-1} \\ n & ; & \frac{1}{n+1} & \leq & \ln(n+1) - \ln(n) & \leq & \frac{1}{n} \end{array}$$

By the sum of the n inequalities; we obtain

$$\ln(n+1) - \ln(n) \leq 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \quad \text{so} \quad \ln(n+1) \leq H_n; \quad n \geq 1 \quad \dots (1)$$

by the sum of the first $(n-1)$ inequalities $\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \leq \ln(n) - \ln(1); \quad n \geq 1$

$$\implies 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \leq \ln(n) + 1; \quad n \geq 1$$

$$\implies H_n \leq \ln(n) + 1; \quad n \geq 1 \quad \dots (2)$$

From (1) and (2), We get $\ln(n+1) \leq H_n \leq \ln(n) + 1; \quad n \geq 1$

c *Limit of the Sequence (H_n)(0.5 pt)*

We have $\ln(n+1) \leq H_n$ and $\lim_{n \rightarrow \infty} \ln(n+1) = +\infty \implies \lim_{n \rightarrow \infty} H_n = +\infty$

d) Let $u_n = H_n - \ln(n), \forall n \geq 1$, * Prove that (u_n) is decreasing.....(01 pt)

$$\begin{aligned} U_{n+1} - U_n &= H_{n+1} - \ln(n+1) - (H_n - \ln(n)) = H_{n+1} - H_n - (\ln(n+1) - \ln(n)) \\ &= \frac{1}{n+1} - \ln\left(\frac{n+1}{n}\right) \end{aligned}$$

From (a) $\frac{1}{n+1} \leq \ln(n+1) - \ln(n)$ so $U_{n+1} - U_n \leq 0$ and (U_n) is decreasing.

* Prove that (u_n) is positive.....(0.5 pt)

$$\text{We have } H_n \geq \ln(n+1) \implies H_n - \ln(n) \geq \ln(n+1) - \ln(n)$$

$$\implies U_n \geq \ln\left(\frac{n+1}{n}\right) > 0 \text{ because } \frac{n+1}{n} > 1$$

so (U_n) is bounded below

* Deduce:(0.5 pt)

because (U_n) is decreasing and (U_n) is bounded below
then (U_n) is convergent

3 Let's consider the sequence (v_n) defined by $v_0 = \frac{3}{2}$ and $v_{n+1} = \sqrt{3v_n - 2}$ for $n \geq 1$.

a) Bounding the Sequence (v_n) by induction.....(01 pt)

For $n = 0$ we have $0 \leq v_0 = \frac{3}{2} \leq 2$ so $P(0)$ is holds

Assume that $P(n)$ is true and Prove that $P(n+1)$ is true.

$$\text{we have } 1 \leq v_n \leq 2 \implies 3 \leq 3v_n \leq 6 \implies 1 \leq 3v_n - 2 \leq 4$$

$$\implies 1 \leq \sqrt{3v_n - 2} \leq 2 \implies 1 \leq v_{n+1} \leq 2$$

then; $\forall n \in \mathbb{N}; 1 \leq v_n \leq 2$

b) Monotonicity and Limit.....(01 pt)

$$v_{n+1} - v_n = \sqrt{3v_n - 2} - v_n = \frac{3v_n - 2 - v_n^2}{\sqrt{3v_n - 2} + v_n}$$

$$\Delta = 9 - 4(-2)(-1) = 1; \quad x_1 = \frac{-3+1}{-2} = 1; \quad x_2 = \frac{-3-1}{-2} = 2$$

v_n	1	2
$v_{n+1} - v_n$	0	+ 0

So (v_n) is increasing.

Since (v_n) is increasing and bounded from above; then it converges and.....(0.5+0.5 pt)

$$\lim_{n \rightarrow \infty} v_n = \lim_{n \rightarrow \infty} v_{n+1} = l \implies l = \sqrt{3l - 2} \implies l^2 - 3l + 2 = 0$$

$$l = 1 \vee l = 2 \text{ since } 1 < v_0 = \frac{3}{2} \text{ and } (v_n) \text{ increasing then } l = 2$$

Exercise 3

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & \text{If } x > 0, \\ e^{x^2} - \cos(x) & \text{If } x \leq 0. \end{cases}$$

- 1 Study the continuity of f on $\mathbb{R}$.
- 2 Study the differentiability of f on $\mathbb{R}$.
- 3 Is f of class C^1 on $\mathbb{R}$? Justify your answer.

Solution 3 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & \text{If } x > 0, \\ e^{x^2} - \cos(x) & \text{If } x \leq 0. \end{cases}$$

- 1 Study the continuity of f on $\mathbb{R}$(01.5 pt)

For $x \in]-\infty, 0[\cup]0, +\infty[$ f is continuous as the product, composition and sum of continuous functions on $\mathbb{R}^*$.

$$* \text{ for } x = 0 : \lim_{x \rightarrow < 0} f(x) = \lim_{x \rightarrow < 0} e^{x^2} - \cos(x) = 0 = f(0) \text{(1)}$$

$$* \lim_{x \rightarrow > 0} f(x) : \text{ we have } -1 \leq \sin(\frac{1}{x}) \leq 1 \implies -x^2 \leq x^2 \sin(\frac{1}{x}) \leq x^2; \quad (x^2 > 0)$$

$$\lim_{x \rightarrow > 0} (-x^2) = \lim_{x \rightarrow > 0} (x^2) = 0 \text{ by the squeeze theorem ; we get ; } \lim_{x \rightarrow > 0} f(x) = 0 \text{ ... (2)}$$

$$(1) \text{ and } (2) \text{ we find } \lim_{x \rightarrow < 0} f(x) = \lim_{x \rightarrow > 0} f(x) = f(0)$$

$$\implies f \text{ is continuous at } x = 0$$

2 Study the differentiability of f on $\mathbb{R}$(0.5+0.5+ 0.5+0.5 pts)

For $x \in]-\infty, 0[\cup]0, +\infty[$ f is differentiable as the product, composition and sum of differentiable functions on $\mathbb{R}^*$.

$$* \text{ for } x = 0 : \lim_{x \rightarrow <0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow <0} \frac{e^{x^2} - \cos x}{x} = \frac{0}{0}$$

$$\text{using l'Hopital's rule ,we get } \lim_{x \rightarrow <0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow <0} \frac{2xe^{x^2} + \sin(x)}{1} = 0 = f'_l(0)$$

$$\lim_{x \rightarrow >0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow >0} x \sin\left(\frac{1}{x}\right)$$

$$-x < x \sin\left(\frac{1}{x}\right) < x; \quad (x > 0) \implies \lim_{x \rightarrow >0} x \sin\left(\frac{1}{x}\right) = 0$$

$$\text{we get } \lim_{x \rightarrow >0} \frac{f(x) - f(0)}{x - 0} = 0 = f'_r(0)$$

we have $f'_l(0) = f'_r(0)$ so f is differentiable at $x = 0$ and $f'(0) = 0$

3 Is f of class C^1 on $\mathbb{R}$ (0.5+0.5+0.5 pts)

$$\text{we have } f'(x) = \begin{cases} 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right); & x > 0 \\ 0 & ; \quad x = 0 \\ 2xe^{x^2} + \sin x & ; \quad x < 0 \end{cases}$$

$$\text{we have } \lim_{x \rightarrow >0} f'(x) = \lim_{x \rightarrow >0} \cos\left(\frac{1}{x}\right) \text{ doesn't exist, so } f' \text{ is not continuous at } x = 0$$

$$\implies f \notin C^1(\mathbb{R})$$